2025 Invitational Tournament - Div C -Georgia

DYNAMIC PLANET

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Instructions:

1. You have 50 minutes to complete this test.
2. Do not write on this test, write on the answer sheet.
3. Question 37 is the first tiebreaker, 38 the second tiebreaker, and 39 the third tiebreaker. Completion time will be used as the final tiebreaker if needed.

Section I: Multiple Choice (3 pts. each)

1. What topographic feature is identified by the area within the red circle on this image below?
   A. Trench
   B. Continental Shelf
   C. Mid-Ocean Ridge
   D. Seamount

2. At what rate are the plates of the Mid-Atlantic Ridge moving apart?
   A. 2.0 cm per year
   B. 2.5 cm per year
   C. 3.0 cm per year
   D. 3.5 cm per year

3. Which of the following statements about turbidites is false?
   A. Sediments are transported and deposited by frictional flow, not by density or tractional flow.
   B. Turbidite deposits typically occur in foreland basins.
   C. Coarser sediments are usually deposited at the bottom, while finer sediments are deposited at the top.
   D. Turbidity currents are often caused by slope failures.

4. The fracture zone is an area of transverse faults around what features?
   A. Deep ocean trenches
   B. Hot spots
   C. Mid-Ocean ridges
   D. Atolls
   E. Guyots

5. When the wind is travelling north, on which of these coasts does downwelling occur?
   A. The east coast of Fiji
   B. The west coast of South Africa
   C. The west coast of Mexico
   D. The east coast of Mozambique

6. What effect do sea walls have on adjacent beaches
   A. Have no effect on erosion
   B. Increase erosion
   C. Decrease erosion
   D. None of the above

7. At the poles, thermocline is
   A. More or less vertical/nonexistent
   B. Extremely shallow
   C. Moderately steep
   D. None of the above

8. A group of students are studying ocean currents at a beach. They notice that the waves are approaching the shore at an angle, moving sand and swimmers steadily *along* the beach. Later, they observe a strong, narrow current pulling water rapidly *away* from the shore through a break in the sandbar.

   Which of the following best matches the two types of currents the students observed?

   A. The first current is a longshore current, and the second is a rip current.
   B. The first current is a rip current, and the second is a longshore current.
   C. Both currents are examples of longshore drift.
   D. Both currents are examples of rip currents.

9. If high tide in Ocean City occurs at 12:15 pm, the next high tide will be at:
   A. 12:40 pm the same day
   B. 12:40 am that night
   C. 12:40 pm the next day
   D. 1:05 pm the next day

10. Which piece of evidence best supports the theory of seafloor spreading?

   A. Fossils of the same species found on different continents
   B. Earthquakes occurring along transform boundaries
   C. The presence of folded mountain ranges along continental margins
   D. Magnetic stripes on the ocean floor that are symmetrical on either side of mid-ocean ridges

Section 2: Matching (2 pts. each)

Match the following photos to the correct depositional landform

a. Spit
b. Barrier Island
c. Lagoon
d. Tombolo
e. Sand Dune

11. 

12. 

13. 

14. 

15. 

Section III: True/False (1 pt. each)

16. Marcet's Principle holds when residence time is lower than mixing time.

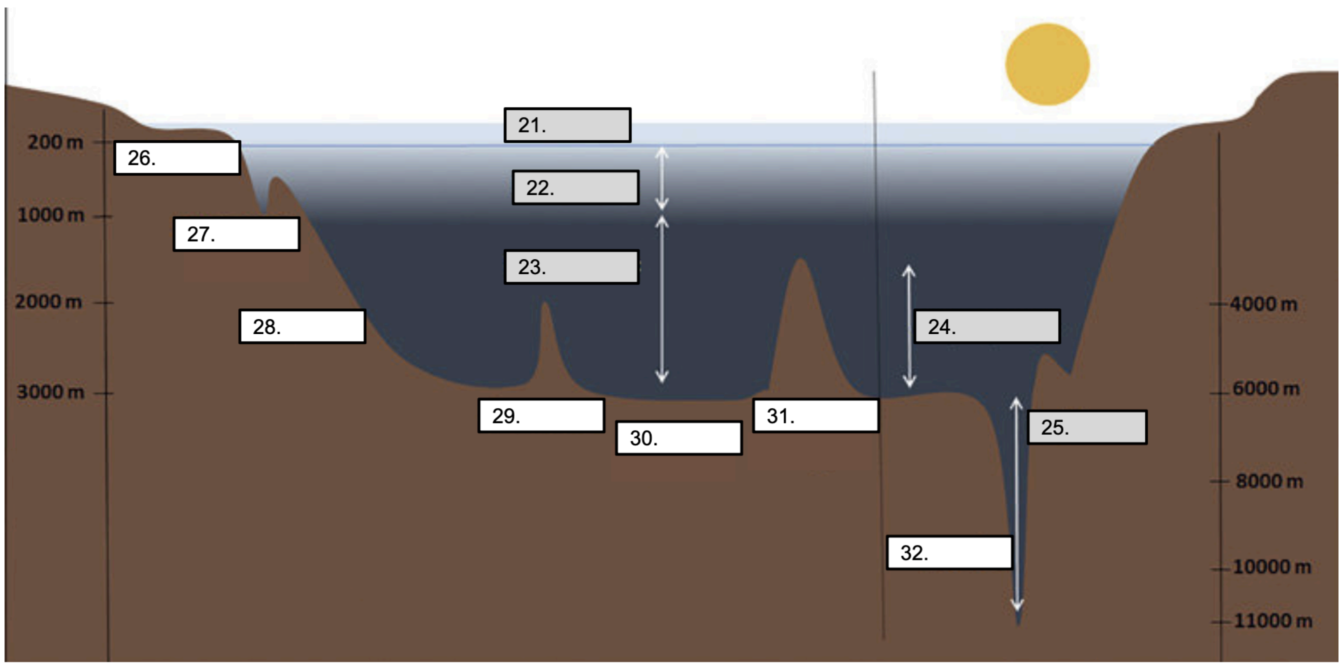
17. The dominant restoring force for a capillary wave is surface tension.

18. Soil contains a greater amount of water than the atmosphere on Earth.

19. Estuaries usually have very high biodiversity because their stable salinity makes it easy for most organisms to survive.

20. If the Greenland ice sheet melts significantly, it could slow down thermohaline circulation by freshening the North Atlantic surface waters.

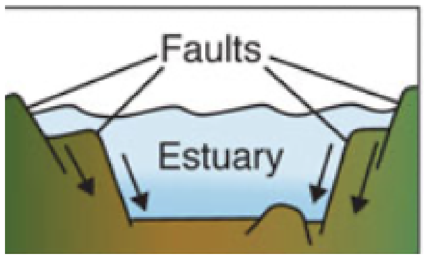
Section IV: Labeling (2 pts. each)



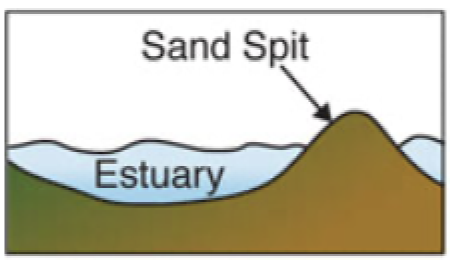
21-32. Label the oceanic zones and seafloor features.

33-35.
Label the landforms pictured below.

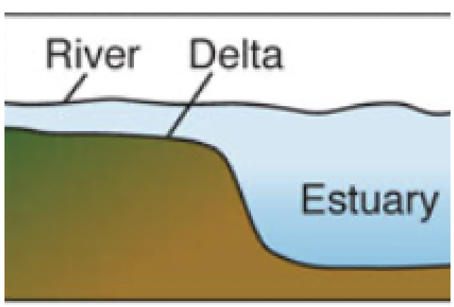
33.



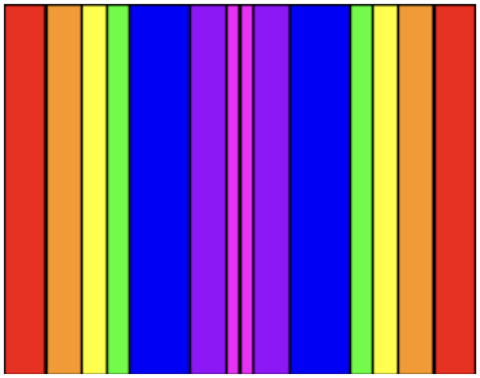
34.



35.



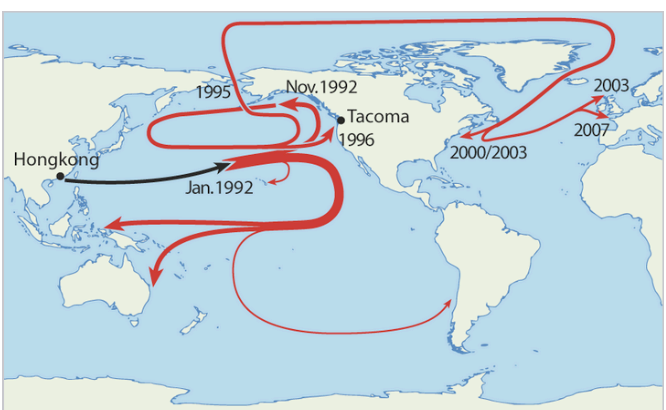
Section V: Short Answer (5 pts. each)



36. Use the above image for the following questions. (Each part is worth 1 point).
    a. Which color strip represents the oldest sea floor?
    b. Which color strip represents the youngest sea floor?
    c. Around which ocean floor feature would you find strips like this?
    d. What causes the formation of these strips?
    e. What plate boundary would create this phenomenon?

37. Explain how Ekman upwelling helps regulate the amount of dissolved oxygen within a few kilometers of the surface.

38. How much do upwelling regions contribute to the world's total fish catch? Give a percentage. **(Tiebreaker)**



39. What event is the map pictured above describing? **(Tiebreaker)**

40. What is the name of the barrier island where the University of Georgia Marine Institute is located? **(Tiebreaker)**